

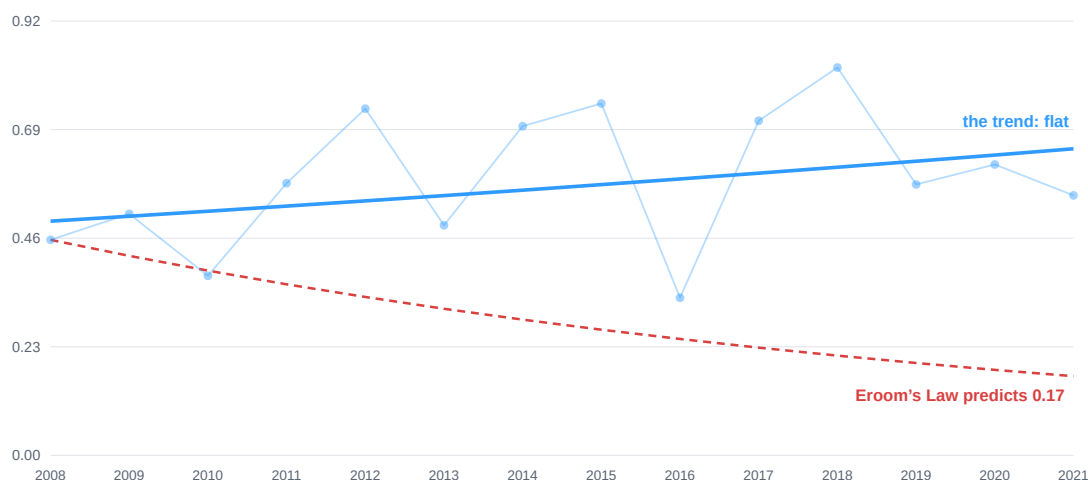
Pharma's most quoted number stopped being true

The US, 2008–2021: Eroom's Law said drug productivity halves every nine years

US new molecular entities approved per billion dollars of US pharmaceutical business R&D, in constant prices.

Red dashed: where the ratio should be if the law still held. Blue: the trend through what actually happened, with the yearly figures behind it.

It did not continue: the ratio is flat to rising, and the law's own rate is rejected in every specification tested.



By 2021 the law implies 0.17 approvals per billion. The figure was 0.55 — more than 3 times higher.

The swings are approvals, not spending: real R&D grows smoothly and accounts for only 10% of the ratio's movement.

This tests only whether the published rate continued. The paper's own 1950–2010 window cannot be re-measured from open data.

Data: FDA Drugs@FDA submission records (new molecular entities) and Eurostat rd_e_berdind2 NACE C21, deflated by the US GDP price index · namikakmandev.github.io/pharma-eroom.html

Scannell et al., Nature Reviews Drug Discovery, 2012

Moore's Law, backwards

Computing power doubles every two years.

New drugs per billion dollars of research **halved** every nine years.
Since 1950.

So they spelled Moore backwards: **Eroom's Law**.

Its data stops in 2010. It is still on slides.

By 2021, the law says

0.17

new drugs per billion dollars.

It was

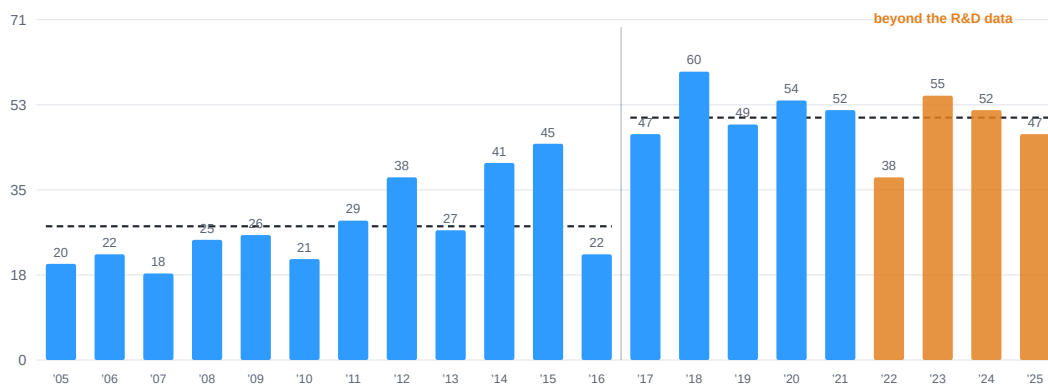
0.55

3 times higher. Rebuilt from the FDA's own submission records against Eurostat's R&D accounts, deflated. The law's rate falls outside the confidence interval in **all 5** US specifications.

The rise happened before AI could have caused it

New molecular entities approved by the FDA each year. The step up is complete by 2017; AI-designed molecules reached the clinic around 2020 and have almost no approvals yet. This does not test what did cause it.

2005–2016: 28 a year on average. 2017–2025: 50. 1.8× more.



The ratio chart stops in 2021 because the R&D series does. Approvals run to 2025; 2026 is a part year at the data cut-off and is left out.

The orange bars are years with approvals but no matching R&D figure, so they cannot enter the productivity ratio.

Data: FDA Drugs@FDA submission records, original applications approved as submission class 7 or 8 · namikakmandev.github.io/pharma-eroom.html

Two things before you quote it

- 1. It is an approvals story.** Spending grew steadily throughout; approvals did the moving. The denominator is only 10% of the ratio's movement, so this is not proof that research got more efficient.
- 2. Date the claim.** The nine-year halving describes 1950–2010. On the years since, it is rejected.

Method, every number and both scripts:

namikakmandev.github.io/pharma-eroom.html

Data: FDA Drugs@FDA submission records; Eurostat rd_e_berdindr2 NACE C21; US GDP deflator. The paper's own 1950–2010 window is not re-measured — no open R&D series reaches back that far.